ONE-PAGE BRIEF

Post-quantum cryptography review: openpgpjs @ 8bc35df48879

Client: OpenPGP.js project — public repository (passive review) · Source: github.com/openpgpjs/openpgpjs, commit `8bc35df48879` · Reviewed: 2026-11-04 20:13 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 2 of 4

post-quantum in use outside the critical paths

FINDINGS

0 Critical **0** High **8** Medium **0** Low **1** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

WHAT MATTERS MOST

Medium QSI-2026-13322	**Quantum-vulnerable X25519: key wrapping and data encryption at rest (5 file(s))** preliminary What to do: Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its...
Medium QSI-2026-13327	**Quantum-vulnerable ECDH: key wrapping and data encryption at rest (1 file(s))** preliminary What to do: Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its...
Medium QSI-2026-13321	**Quantum-vulnerable ECDSA: role not determined (7 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium QSI-2026-13325	**Quantum-vulnerable ECDH: role not determined (2 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in...

4 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium key wrapping and data encryption at rest	X25519, ECDH	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may...
Medium role not determined	ECDSA, X25519, ECDH, RSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Static review of source code and manifests; files analysed: 108. No requests were sent to the owner's systems.

Full report, content SHA-256: `cba05e60ddad459b1d599134c411395e900bf2b3adb93f02cfda6d12c3fc69d5`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.